# Bacterial degradation of Di-(2-ethylhexyl) phthalate (DEHP) and its *insilico* analysis of interaction with Carboxylesterase Est2

*A dissertation submitted for the partial fulfilment of*
*the award of degree*
*of*

**INTEGRATED MASTER OF SCIENCE**

*in*

**MICROBIOLOGY**


*Submitted by*

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**(2019IMSMB014)**

*Under the Supervision of*

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(A Central University Established by the Act of Parliament)

# CERTIFICATE

This is to certify that the dissertation entitled "**Bacterial degradation of Di-(2-ethylhexyl) phthalate (DEHP) and its *insilico* analysis of interaction with Carboxylesterase Est2"** has been submitted by Mr. **Piyush Gurjar** bearing Enrollment No. **2019IMSMB014,** a student of Integrated M.Sc. Microbiology, Dept. of Microbiology, Central University of Rajasthan, India, under my supervision. The work presented in the thesis is original and has not been submitted to any other university or institute for the award of any other degree/diploma.

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# DECLARATION

I, **Piyush Gurjar**, declare that the Dissertation Project entitled "**Bacterial degradation of Di-(2-ethylhexyl) phthalate (DEHP) and its *insilico* analysis of interaction with Carboxylesterase Est2**" comprise the result of dissertation work carried out by me under the supervision of **Dr. L Paikhomba Singha** for the award of an Integrated M.Sc. degree in Microbiology, Department of Microbiology, School of Life Science, Central University of Rajasthan.

I further declare that the presented work is within the permissible range of 10% similarity certified by Turnitin software and the report is enclosed on the last page of dissertation for verification.

Date: 03rd June 2024

Place: Ajmer

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# ABSTRACT

Di-(2-ethylhexyl) phthalate (DEHP) is a widely used plasticizer present in numerous consumer products, such as medical devices, packaging materials, and household items. Due to its non-covalent attachment to plastics, DEHP can easily leach into the environment, contaminating air, water, and soil. This widespread contamination poses serious health risks due to DEHP's endocrine-disrupting properties and potential toxicity to both wildlife and humans. This dissertation focused on the microbial degradation of DEHP, with particular attention to the bacterium *Acinetobacter oleivorans* DR1, investigated the molecular interaction of DEHP with the enzyme this bacterium had showed 82.8% degradation of DEHP at 72 hrs of incubation. Carboxylesterase Est2 using *insilico* methods and how biodegradation efficacy of the enzyme Est2 can be increased. This enzyme was selected due to its high sequence similarity (82%) to the known DEHP-degrading enzyme M673. Molecular docking studies were used to model the binding interactions between DEHP and Carboxylesterase Est2, identifying key residues involved in substrate recognition and catalysis. The in-silico analysis provided detailed insights into the enzyme's active site and suggested potential modifications to enhance its catalytic efficiency. The results of this study have significant environmental implications, suggesting that microbial degradation, particularly by *Acinetobacter oleivorans* DR1, could be a viable strategy for bioremediation of DEHP-contaminated sites. The present study will be useful in understanding DEHP degradation and provides a foundation for developing bioremediation strategies to mitigate the environmental impact of plasticizers. The integration of experimental and computational approaches offers a comprehensive framework for addressing the challenges associated with DEHP contamination and sets the stage for future research in environmental microbiology.

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# ABBREVIATIONS

PAE’s : Phthalic Acid Esters

Thr: Threonine

Phe: Phenylalanine

Ser: Serine

Ala: Alanine

Ile: Isoleucine

Arg: Arginine

Asp: Aspartic Acid

Leu: Leucine

His: Histidine

Glu: Glutamine

IUPAC: International Union of Pure and Applied Chemistry

DEP: Diethyl Phthalate

DBP: Dibutyl Phthalate

MEHP: Mono ethylhexyl phthalate

BTEX: Benzene, Toluene, Ethylbenzene and Xylene

PBS**:** Phosphate Buffer Saline

NCBI: National Centre of Biotechnology Information

JGI-IMG: Joint Genome Institute-Integrated Microbial Genomes and Microbiomes

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# INTRODUCTION

Phthalates are the esters of the *o*-phthalic acid (Singh et al., 2017). They are primarily used as the plasticizer, giving a different strength, flexibility, and longevity to the plastics. Phthalates acid Esters (PAE's) are used in toys, bottles, medical devices, cosmetics, and buildings and have many more applications (Meeker et al., 2009). We are highly dependent on plastics and indirectly on phthalates as they give an edge difference to the plastics' properties. Phthalate exposure is one of the major concerns today as we are highly using plastic products in every sphere be it in schools, homes, or construction sites (Wang and Qian, 2021).

There are different kinds of phthalates based on their chemical and physical properties, but there are some kinds of phthalates that are in high use such as Bis (2-ethyl hexyl) phthalates or Di ethyl hexyl phthalate (DEHP), Di-n-butyl phthalate, Di-n-octyl phthalate (Barakat and Ko, 2018).

DEHP is the most common phthalate which is used in the plastic industry. It is a colorless, odorless liquid with the molecular formula of $C_{24}H_{38}O_4$ and a molecular weight of 390.56 g/mol. It is non-covalently bonded to the plastics (Barakat and Ko, 2018) (Rowdhwal and Chen, 2018). DEHP is used in a variety of products including medical devices, such as intravenous (IV) bags and tubing, umbilical artery catheters, blood bags and infusion tubing, enteral nutrition feeding bags, nasogastric tubes, and peritoneal dialysis bags, and is also utilized in manufacturing a wide variety of consumer products, such as packed food and beverages; soft plastic products, such as toys and infant products, furniture upholstery, mattresses, wall coverings, floor tiles, and vinyl flooring; and cosmetics and personal care products to carry fragrances (Rowdhwal and Chen, 2018).

Due to this wide use and non-covalent bond between DEHP and plastics, the DEHP is found in air, water, at the production site, and in every possible environment (Rowdhwal and Chen, 2018). This led to a high exposure rate of the DEHP in all animals including humans.

DEHP has different kinds of risks as it is carcinogenic when some phthalates including DEHP were given to rats their liver suffered an increase in liver weight, higher liver enzyme levels, and in certain cases tumors (Nikonorow et al., 1973). DEHP also has detrimental effects on reproductive health studies done on rats when rats were supplemented with 0.35% DEHP for one year in their diet it caused an increase in their kidney weight and a decrease in their body weight (McKee et al., 2002). Anomalies in neurodevelopment because of DEHP exposure have also been reported as they affect a child's mental and motor development. As Whyatt et al., 2012, reported that the child's mental and motor development especially of girl child is affected by the metabolites of DEHP. Endocrine disruption is also one of the major effects that DEHP has on the animal body (Pocar et al., 2023).

DEHP is not only detected in the terrestrial ecosystem but also in aquatic ecosystems. During the production stage, the waste is mainly disposed to the moving water, and it is the primary exposure of phthalates to the water streams after production, these compounds find their way to water as the runoff carries all pollutants and effluents (Zhang et al., 2021).

The various effects of DEHP in aquatic ecosystems on different aquatic organisms include Endocrine disruption (Forget-Leray et al., 2005), reduced oocyte fertilization rates (Ye et al., 2014), decreased sperm production, velocity, and motility (Golshan et al., 2015), etc.

Microbial degradation is the most important route for DEHP remediation, due to its cost-effectiveness, less harmfulness, ecofriendly nature, versatileness, faster than hydrolysis and photolysis, and long-term effectiveness. Biodegradation can be affected by several factors including temperature, pH, nutrient availability, and salinity (Chang et al., 2022).

The microbial degradation of DEHP is done by the following steps, first, the DEHP is degraded to MEHP and then to the phthalic acid (PA), which is followed by the two different mechanisms of the degradation including the aerobic and anaerobic degradation mechanism. The aerobic degradation is much faster than the anaerobic degradation of the DEHP. The aerobic degradation of PA is followed by the formation of protocatechuate and then to the oxaloacetate and pyruvate, and in anaerobic degradation phthalic acid is decarboxylated to benzoate and then to acetate. Further degradation is done by the TCA cycle (Eaton and Ribbons, 1982), (Kleerebezem et al., 1999), (Liu and Chi, 2003).

Several In-Silico studies have been conducted for molecular insights and interactions between different enzymes and DEHP (Singh et al., 2017), but to date, no study has focused on *Acinetobacter oleivorans* DR1, and to hypothesize the increase in the efficiency of the enzyme responsible for the DEHP degradation. For the Insilco studies, we have chosen the Carboxylesterase Est2 enzyme from the DR1 as it is found 82% similarities with the reported DEHP degrading enzyme M673 from the *Acinetobacter* sp. M673 (Li et al., 2022).

carboxylesterase is an HSL class hydrolase having a conserved pentapeptide ($GX_1SX_2GG$) and catalytic triad (S-H-D/E) where serine Or acts as a nucleophilic center for hydrolyzing Ester, amide, or the thioester bonds (De Simone et al., 2004).

The study focuses on the growing concern over environmental toxins, particularly the widespread presence and impact of Di-(2-ethylhexyl) phthalate (DEHP), a common plasticizer. Given the harmful effects of DEHP on human health and the environment, the study aims to explore biodegradation as a viable solution for mitigating these risks. The study underscores the importance of innovative bioremediation strategies that combine microbiological techniques with computational tools to address the challenges posed by persistent environmental contaminants.

## OBJECTIVES

1. Invitro screening of DEHP degrading bacteria and annotation of DEHP degrading genes in the bacterial genome.
2. Molecular docking of the Carboxylesterase Est2 with the DEHP and mutational analysis of the active site for the improvement of binding efficiency.

# REVIEW OF LITERATURE

Plastics have been very useful these days since its inception in 1906 (Swallow, 1951). We plastics in our home, offices, schools, etc., they give us a simple life, but they also have harmful effects on us which become one of the major concerns of our planet and ecosystem (Thompson et al., 2009). We are constantly getting exposed to plastics through various sources including food, leach-out compounds from medical equipment, plastic packaging, plastic wares, dust containing microplastics, etc., even cosmetics have plastic contamination, and the generated plastic is not recycled at the same pace (Rahimi and García, 2017).

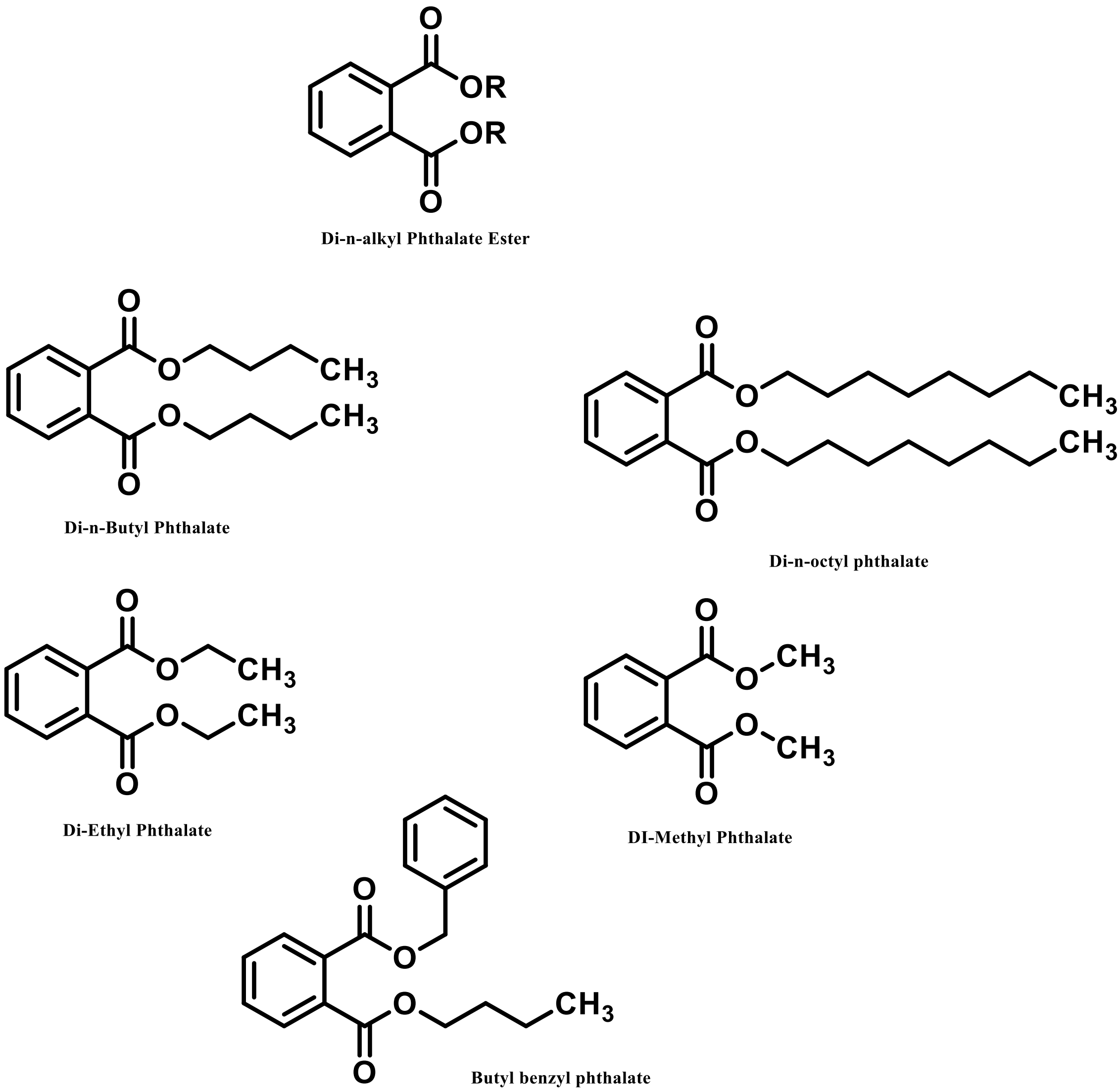


*Figure 1. Different PAEs structures.*

We are highly dependent on plastics in our daily lives, but plastic exposure for long periods has various severe risks like endocrine disruption, cancer, heart-related problems, respiratory inflammation, reproductive toxicity, etc. (Xia et al., 2018), and the most concerning in the plastic is the additives like Bisphenol-A, Phthalates, and various Polyaromatic hydrocarbons.

Phthalates are dialkyl or alkyl aryl Esters of phthalic acid that are added to plastics to increase their flexibility, longevity, durability (Ren et al., 2018), and are mainly added to the Poly-Vinyl Chloride (PVC) which is in about 30-35% (w/w). Phthalates are not only substances that are endocrine disrupters, carcinogenic, toxic to reproductive health but they also affect the lower age group more as compared to the adults, especially during early growth. Various experimental and in-silico research have highlighted the toxic effects of phthalates (Heudorf et al., 2007); (Kimber and Dearman, 2010).

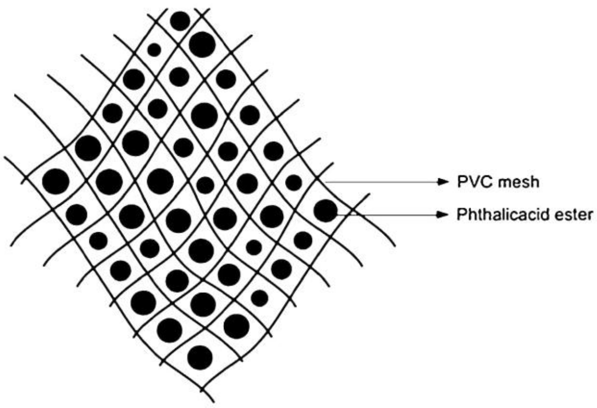


*Figure 2. Phthalic acid Ester in plastics (Benjamin et al., 2015)*

The United States Environmental Protection Agency (USEPA) and European Union (EU) have placed six dialkyl (Fig 1. and 4) phthalates in the priority pollutants list and are monitored for the well-being of mankind (Net et al., 2015) (Kaur et al., 2017); (Kumari and Kaur, 2021). Phthalic acid Esters have a life of hundreds of years (Huang et al., 2021) and phthalates are non-covalently bonded to the plastic causing them to be easily released in the environment from plastic products (Hahladakis et al., 2018); (Takada and Karapanagioti, 2019). Di (2-ethylhexyl) phthalate (DEHP) is the most used phthalates that make plastics more flexible and longevity, it has the IUPAC name "bis(2-ethylhexyl) phthalate". DEHP has a global market of US$ 9.1 bn in 2022 and it is expected to increase by 2031 to US$ 13.1 Bn ("Di (2-ethylhexyl) Phthalate Market," n.d.). DEHP is produced by the reaction of phthalic anhydride (1) with excess 2-ethylhexanol (2) in the presence of an acid catalyst.

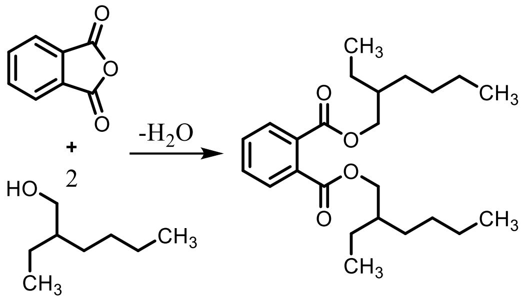

*Figure 3. Production of DEHP*

It is colorless, viscous, and hydrophobic chemical, which consist of 24 carbons, having chemical formula $C_{24}H_{34}O_4$ and molecular weight of 390.56 g/mol (Rowdhwal and Chen, 2018).

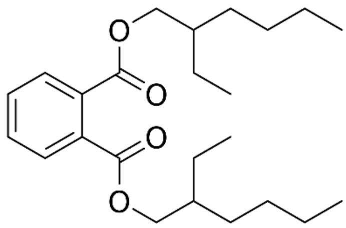

*Figure 4. DEHP structure.*

## Health effects of DEHP on Human body:

***Neurotoxicity***

DEHP affects the brain cells and causes severe teratogenic anomalies and hinders the neural development. As investigated by Xu et al., 2007, DEHP have severe effects on the rats brains in both gestational and post-natal period.

***Endocrine Toxicity***

DEHP is known as an endocrine disrupter (ER), which alters the function of the endocrine system and have deleterious health effects to the organism (Fig.4). Martinez-Arguelles et al., 2009 and Martinez-Arguelles et al., 2011 have investigate that the aldosterone and testosterone concentrations in male rats decreases up to about 50% due to DEHP exposure. Not only DEHP but also it's metabolites such as MEHP causes the endocrine toxicity as reported by Zhai et al., 2014 and Jia et al., 2016 on zebrafish reported that DEHP does not only alter the thyroid levels but also affect their synthesis, regulation and action.

***Renal toxicity***

DEHP is nephrotoxic and it lowers the weight of the kidney (David et al., 2000a). DEHP exposed male and female rats also show pigmentation when exposed to an higher amount (David et al., 2000b). In an investigated by Faouzi et al., 1999 post dialysis the amount of DEHP in body is increased by 3.6 to 59.6 mg as the dialysis tube is made up of plastic.

***Other Effects of DEHP Exposure***

DEHP have various severe effects on the genital development, reproductive system, and compared to adults children are more vulnerable to DEHP exposure and have side effects due to it's exposure (Heudorf et al., 2007).

As investigated by (Huang et al., 2012) invivo exposure of DEHP in rats led to reduction in fertility, decline in developmental potentiality, and he reported that treatment of rats for 1 month results in increase in the mutations in DNA.

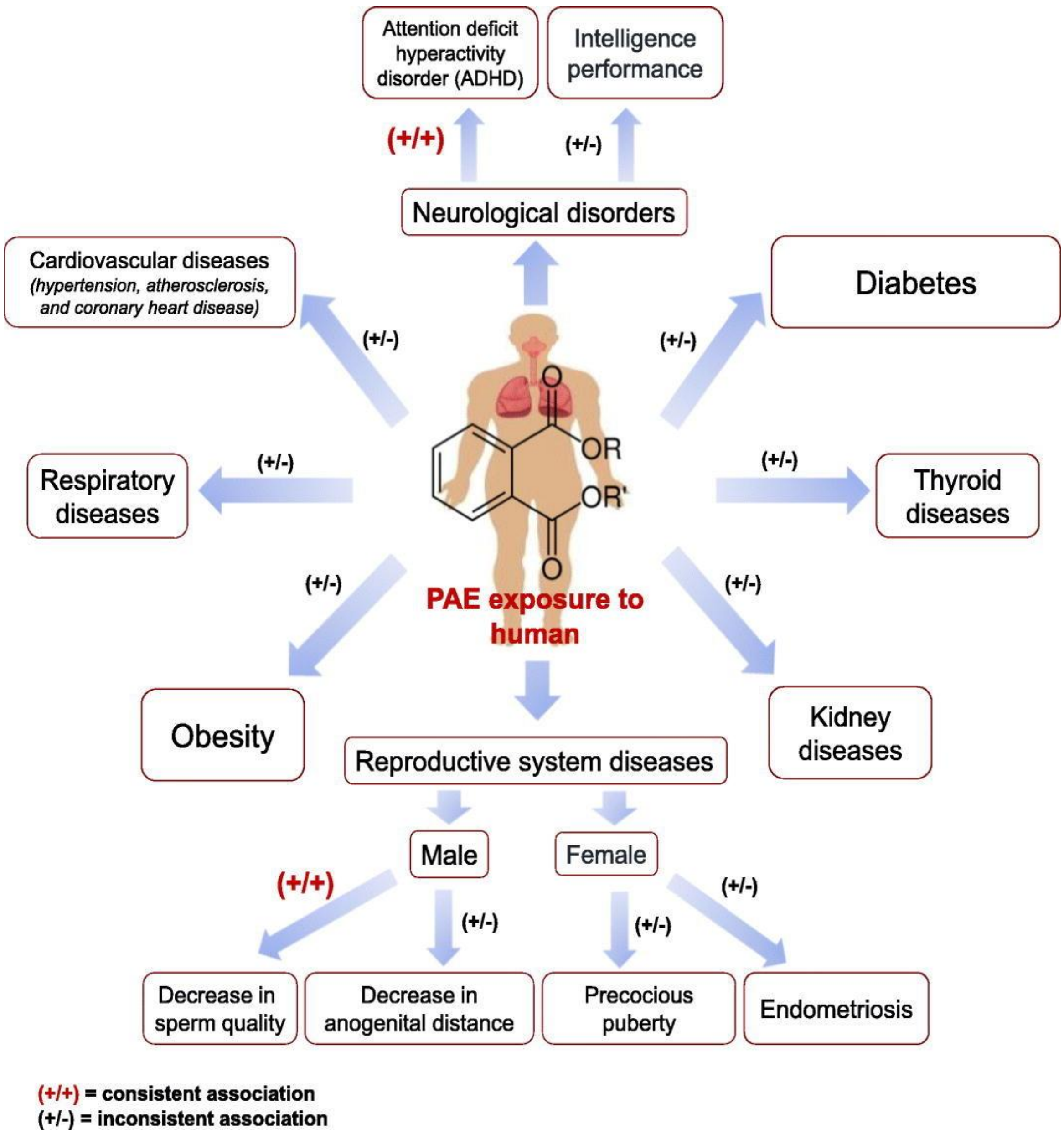


*Figure 5. DEHP effects the human body. (Chang et al., 2021)*

## Different methods of phthalate remediation

Phthalates including DEP, DBP, MEHP, and DEHP are used in many products like medical devices, furniture materials, water pipes, thermal papers, toys, plastic bottles, mattresses, wall coverings, cosmetics, fragrances, etc (Taniguchi et al., 2019). Due to the high use of the DEHP and other phthalates, they are found in every sphere including air, water, soil, etc., their contamination poses a significant challenge, and there is an urgent need for it's remediation. For this there are different methods available for the remediation the phthalates like chemical hydrolytic degradation, photolysis, sonolysis, Biodegradation, etc.

***Chemical hydrolysis***

Hydrolysis is the chemical reaction in which a water molecule breaks the chemical bonds of any compound. It is the major or we can say dominant process which degrades the phthalates in the landfill sites (Huang et al., 2013). As the landfill depth increases so the hydrolysis of phthalates increases as the temperature increase which increases the rate (Schwarzbauer et al., 2006).

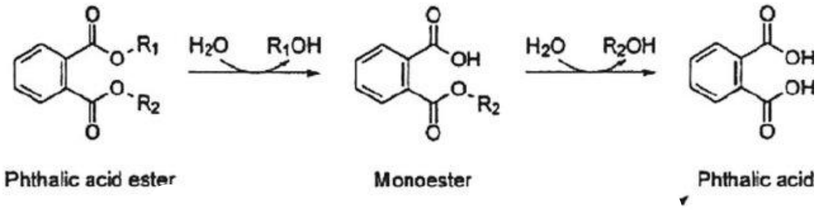


*Figure 6. Transformation of phthalate to phthalic acid.*

There are different factors that affect the hydrolysis process such as pH (acidic and basic conditions) for the hydrolysis (Huang et al., 2013), temperature (Schwarzbauer et al., 2006)

***Photolysis***

In this, the phthalates are transformed by using two different mechanisms first by ozonation and second by using the reactive oxygen species such as hydroxyl free radicals which are generated from the decomposition of ozone in an aqueous solution (Hoffmann et al., 1995); (Staples et al., 1997).

Yu et al., 2019 investigate the photolysis of DEHP under simulated sunlight in the natural water photoreactive constituents and also investigated the presence of various constituents with varying concentrations of the nitrate ($NO_3^-$), ferric ion ($Fe^{3+}$), Fulvic acids (FA) and he also studied the effect of chloride ion ($Cl^-$) in presence of FA, resulted that the photolysis of DEHP is increases with the increase in the concentration of the $NO_3^-$ and $Fe^{3+}$ but with the increase in the concentration of the FA photolysis have a inhibitory effect on the DEHP photolysis. But when the FA in low concentration and $Cl^-$ coexists it significantly increases the photolysis process.

***Sonolysis***

Sonolysis is the method of the breakdown of chemical compounds using ultrasonic waves i.e. of frequency above 20kHz. (Yim et al., 2002) investigated the degradation of PAEs such as dimethyl phthalate (DMP), diethyl phthalate (DEP), and dibutyl phthalate (DBP) by sonochemical action with the ultrasonic frequency of 200kHz, this study resulted that the sonochemical degradation of phthalates follows pseudo-first-order kinetics and that over the pH 11 degradation rate was accelerated because of abundant of $OH^-$ i.e. increased hydrolysis. Psillakis et al., 2004 also investigate the sonolytic degradation of the six-phthalate including DEHP at 80kHz frequency and found that efficient degradation of phthalates occurs in time band of 30 to 60 mins.

***Biodegradation***

Biodegradation is one of the best options available for remediation of the DEHP and other leachout compound as it is cost-effective, generate less or no byproducts, highly efficient. Biodegradation is an environment-friendly method for the remediation of leachout compounds (Ghosh and Sahu, 2022).

Different microbial species have been reported for the degradation of the DEHP with varying efficiency at different time (Table 1) such as *Pseudomonas spp., Gordonia spp., Arthrobacter spp., Acinetobacter spp., Micrococcus spp., Rhodococcus spp., Microbacterium spp., Bacillus spp., Flavobacterium spp., etc.*

The biodegradation mechanism of the PAEs is relatively same in many bacterial species, as the first step for the degradation is the conversion of the dialkyl ester is to mono ester and then to phthalic acid which will degrade in two mechanisms i.e. aerobic and anaerobic degradation, it is shown Fig 7 for both mechanisms of the phthalate degradation. Most of the microbes involved in phthalate degradation are aerobic, but there are also anaerobic microbes involved like *A. aromaticum* EbN1 (Boll et al., 2020)*, T. chlorobenzoica strain 3CB-1, Azoarcus evansii strain* KB740*, A. aromaticum strain EbN1, T. aromatica strain* K172*, Azoarcus buckelii strain* U120 *and Azoarcus sp.* CIB (Ebenau-Jehle et al., 2017).

*Table 1 Degradation of DEHP by different microbial species*

| Species | Isolated from | Concentration (mg/L) | Performance | References |
|---|---|---|---|---|
| *Sphingomonas* sp. DK4 | Petrochemical waste | 5 | 90% degraded in 5 days | (Chang et al., 2004) |
| *Pseudomonas fluorescens* FS1 | Activated sludge | 50-400 | 20% degraded in 3 days | (Zeng et al., 2004) |
| *Acinetobacter lwoffii* | River water | 20 | 20% degraded in 5 days | (Hashizume et al., 2002) |
| *Mycobacterium* sp. NK301 | Garden soil | 1000 | 98% degraded in 21 hrs | (Nakamiya et al., 2005) |
| *Corynebacterium* sp. O18 | Activated sludge (petrochemical wastewater) | 5 | 57% degraded in 5 days | (Chang et al., 2004) |
| *Gordonia polyisoprenivorans* G1 | Soil | 100 and 243mg/kg in soil | 50% degraded in 2 days and 50% degraded in 3 days | (Chao and Cheng, 2007) |
| *Enterobacter* spp. Strain YC-IL1 | Contaminated soil | 400 | Complete degradation in 7 days | (Lamraoui et al., 2020) |
| *Rhodococcus* G2 and G7 | Soil | 100 | 98.4% degradation in 3 days by G2 and 91.7% by G7 in 5 days | (Chao and Cheng, 2007) |
| *Rhodococcus rhodochrous* ATCC 21766 | Soil | 9360 | 67% degradation in 200 hrs | (Nalli et al., 2006) |
| *Rhodococcus erytropolis* S1 | Activated sludge | 1500-3000 | Complete degradation in 3 to 5 days | (Kurane, 1997) |

| | | | | |
|---|---|---|---|---|
| *Bacillus* sp. S4 | Sludge | 1000 | Degradation constant k=0.081 | (Chang et al., 2007) |
| *Bacillus subtilis* No. 66 | Soil | 3900 | 81.6% degradation in 5 days | (Quan et al., 2005) |
| *Bacillus brevis* | Soil | 1000 | | (Juneson et al., 2001) |
| *Pseudomonas* sp. PKDM2 *Pseudomonas* sp. PKDE1 *Pseudomonas* sp. PKDE2 | Plastic contaminated soil | 500 | Complete degradation in 40-44 hrs | (Singh et al., 2017) |
| *Nocardia asteroides* LMB-7 | Electronic waste soil | 200 | 98.9% degradation in 24hrs | (Chang et al., 2022) |
| *Rhodococcus* sp. strain WJ4 | Soil sample | 200 | 96.4% in 7 days | (Wang et al., 2015) |
| *Fusarium culmorum* | Mixed pulper wasters | 1000 | 98% in 6 days | (Yang et al., 2018) |
| *Agromyces* sp. MT-O | Soil sample | 200 | 100% degradation in 7 days | (Zhao et al., 2016) |
| *Gordonia alkanivorans* YC-RL2 | Petroleum-contaminated soil | 800 | 94.6% in 7 days | (Nahurira et al., 2017) |
| *Microbacterium* sp. CQ0110Y | Activated sludge | 1350 | Complete degradation in 10 days | (Chen et al., 2007) |

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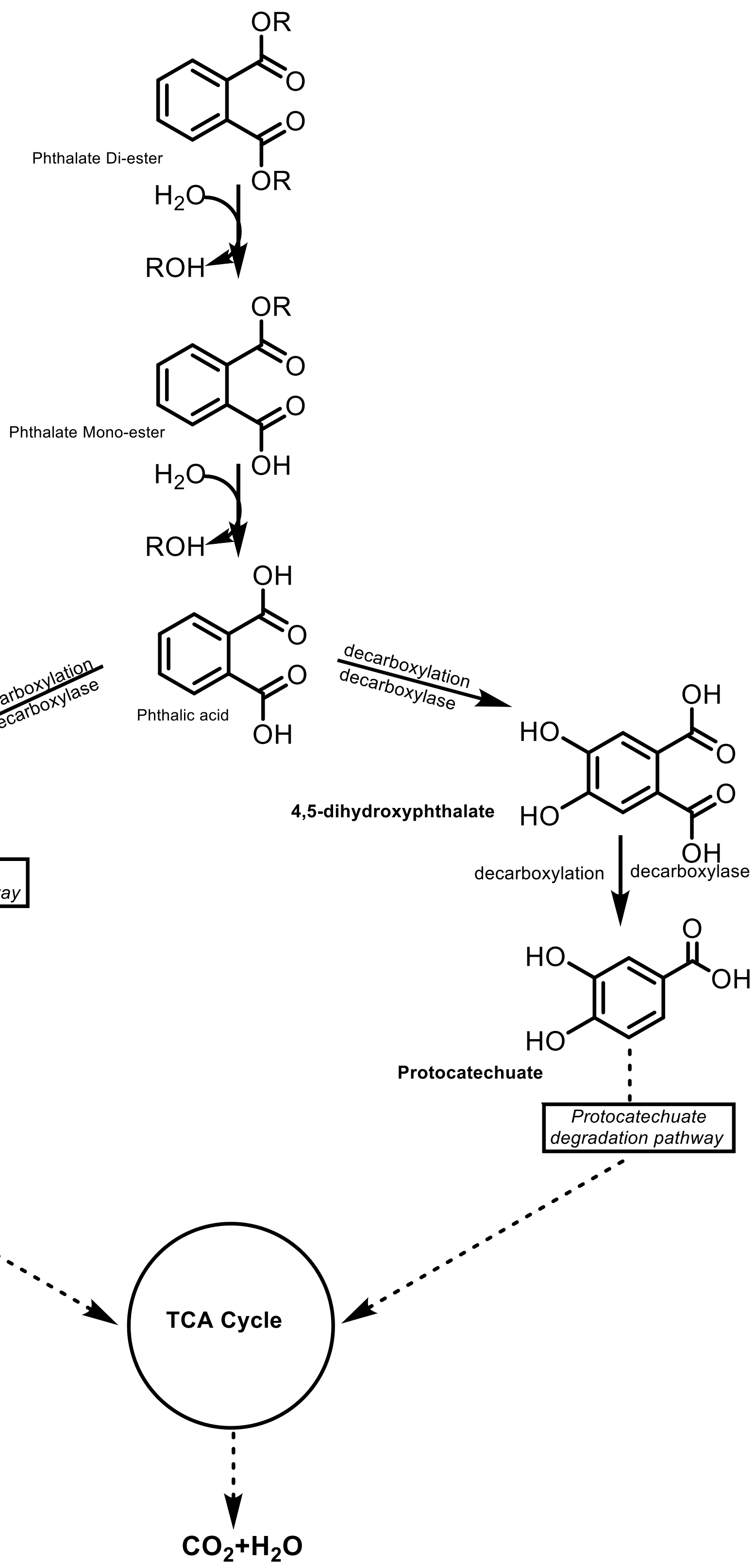



*Figure 7. Degradation of DEHP*

## Molecular Docking

Biodegradation is an Interested research area these days, there are around 16,80,000 journals are available on the google scholar, but less are focusing on the In-Silico approach for the same topic.

There are various environmental pollutants including the polyaromatic hydrocarbons, polychlorinated hydrocarbons, BTEX, and Phenols, derivatives of these groups are not only carcinogenic but also teratogenic, cytotoxic, and mutagenic (Esteve-Turrillas et al., 2007); (Robertson and Ludewig, 2011); (Singha et al., 2018); (Singha and Pandey, 2020); (Mustafa, 2024).

Molecular docking (MD) (Morris and Lim-Wilby, 2008) is a structure-based method which predicts the binding conformation and binding affinity between the molecules i.e. the receptor and ligands. MD found its relevance in medical biology and drug discovery, but nowadays MD is also used in the bioremediation. MD is a very convenient, less time consuming and low-cost method to understand the reaction mechanism of proteins or enzymes with ligands with a high accuracy.

There are different types of bonds or affinities between receptor-ligand, which can be ionic interactions, hydrogen bonding, hydrophobic interaction, and Van der Waals force, etc. these interactions have different binding affinity and form between different atoms of the receptor and ligand (Du et al., 2016).

## Molecular Docking Applications in Biodegradation

The application of the MD in biodegradation research has grown in significance. It makes it possible to choose the best appropriate receptor by predicting the ligand's binding affinity with different receptors. Finding efficient biodegradation routes can be streamlined by using laboratory trials to confirm these *insilico* predictions.

Environmental pollutants including BTEX (Navarro Amador et al., 2018), phenols (Stefani, 2016), phthalates (Singh et al., 2017), are hydrophobic. MD capable to able to forecast the requirement for surfactants by certain microbial species to increase the bioavailability of these pollutants and hence facilitate their breakdown (Zhang et al., 2012).

By examining the conformations of receptor-ligand interactions, molecular docking (MD) is also been useful in forecasting degradation pathways. Singh et al., 2017, for example hypothesised the mehpH mediated degradation mechanism of MEHP, in which the serine residue functions as nucleophile to aid in the breakdown of MEHP into phthalic acid.

## Computational modelling for enhancement of enzyme activity

Biodegradation of environmental pollutants is a hot topic of research nowadays. But due to the different properties and higher chains of the pollutants it is very time taking, and vigorous (Hickey, 2021). For this genetic engineering is chosen as one of the ways to increase their activity towards the pollutants. But genetically engineered bacteria generally have low genetic stability and have biosafety problems due to the difficulty of stably retaining the recombinant plasmids (LaPara et al., 2002). Here MD plays an important role by allowing us to first predict the best ligand or enzyme sequence, which then guides subsequent laboratory research.

Li et al., 2019, studied the Naphthalene dioxygenase (Fig. 9) which is an key metabolic enzymes that degrades the aromatic hydrocarbons (Ensley and Gibson, 1983) and they also studied how it's efficiency can be increased using the computational method, in their study they mutated the hydrophilic interacting amino acids with the pollutants with all the nine hydrophobic amino acids and compared their docking scores, rate constant and energy barriers using the SYBYL-X 2.0, KisTheIP and Gaussian tools, respectively. They hypothesized that binding efficiency and rate of reaction can be increased by changing the hydrophilic amino acid residues by the hydrophobic amino acid for the hydrophobic environmental pollutants.

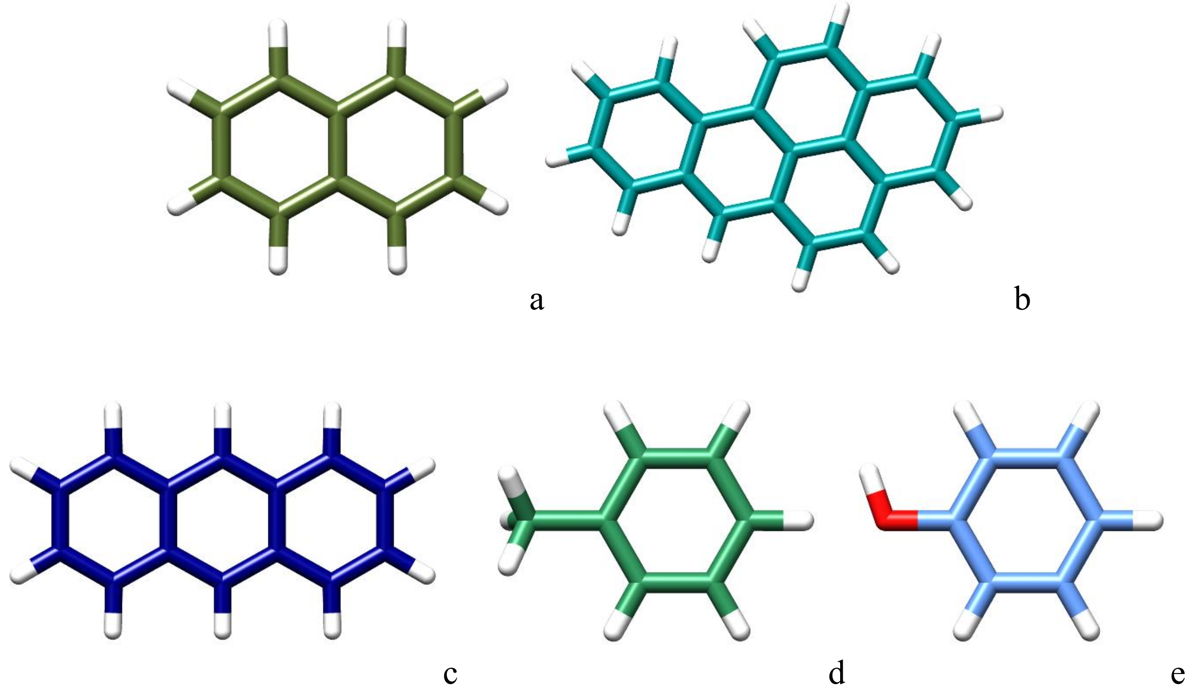



*Figure 8. Structure of environmental pollutants a. Naphthalene, b. Benzo(α)pyrene, c. Anthracene, d. toluene, e. Phenol.*

*Figure 9. Nitrogenase Dioxygenase enzyme.*

# MATERIALS AND METHODS

## Chemicals and Media

DEHP is procured from Otto chemicals with 98% purity (B-1814b), bushnell hass broth, bushnell hass agar, LB broth (M-1245), LB agar (M-1151), nutrient broth (M-002G), nutrient agar (M-001), ethyl acetate (AS-050), all were procured from Hi-Media chemicals, n-hexane (542820) from central drug house of AR grade.

## Bacterial strains and culture conditions

The pure cultures of five bacterial species *Acinetobacter oleivorans* DR1*, Arthrobacter* sp. CAS41-1, *Klebsiella* sp. AWD5, were procured from the ETPB (Enzyme Technology and protein bioinformatics) laboratory, Banaras Hindu University, India. Using plate streak method, quadrant streaking was performed to obtain isolated colonies of bacterial isolates. The different bacterial species were maintained using LBA (Luria Bertani Agar) and stored in glycerol stock at -20 $^{o}$C.

## Screening for Biodegradation

These 3 cultures were grown with DEHP as the sole carbon source in the bushnell hass agar (BHA) plates as the primary screening. After this primary screen, the strains were cultured overnight in LB broth, then it is centrifuged for 10 mins at 8000 rpm at 4 $^{o}$C after that the cell pellet was washed twice with PBS and suspended in PBS with 7.2 pH to make a suspension in PBS of 0.4 $OD_{600}$.. These were inoculated in 5 mL bushnell hass broth (BHB) with DEHP as the sole carbon source and non-inoculated BHB supplemented with DEHP was used as control sample, and all the experiments were performed in duplicates. Best strain that grows on DEHP was selected as the *Acinetobacter oleivorans* DR1 and it is used for further experiments.

## Analysing the DEHP biodegradation

UV-Vis spectrophotometer Shimadzu (UV 2600) was used for the detection of biodegradation and measurement of DEHP amount. For extraction of the DEHP from the culture, Liquid-Liquid extraction method was used, in brief the cultured broth was taken, and equal amount of ethyl acetate was added, then vortexed for 30 seconds, the organic phase was extracted in separate test tube. The optical density of the extract was measured with the UV-Vis spectrophotometry with scan method.

## Congo red test

*Acinetobacter oleivorans* DR1 was tested for the exopolysaccharide with the Congo red dye test Congo Red Agar media consists of BHI broth, 1.5% agar base, and 0.8 g/L Congo Red indicator. Congo Red indicator was prepared as concentrated liquid apart of other media constituents and autoclaved at 121 ºC for 15 minutes and then added into cooled agar in 55 ºC. Then *Acinetobacter oleivorans* DR1 was inoculated in the plates and incubated for 24 hours in 37 ºC. Positive result was shown by black coloured colonies. Negative result was shown by pink colour on colonies.

## Homology search of the DEHP degrading enzyme in the genome of *Acinetobacter oleivorans* DR1

Sequence of M673 an enzyme that is reported for the degradation of PAEs (Wu et al., 2013) from the *Acinetobacter* M673 was taken from NCBI ( https://www.ncbi.nlm.nih.gov/ ) and this sequence was searched in the *Acinetobacter oleivorans* DR1 using JGI-IMG find gene tool (https://img.jgi.doe.gov/) and 5 enzymes with different sequence lengths and identities were found (Table 2).

## DEHP as a Ligand

3D structure of the DEHP was retrieved from the Chem Spider (https://www.chemspider.com) in mol format and it is converted to the .pdb format and then to .pdbqt using Open babel software (O'Boyle et al., 2011).

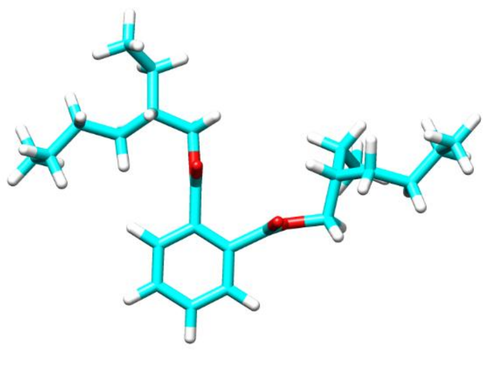


*Figure 10. 3D structure of DEHP.*

## Structural validation of Carboxylesterase Est2

Structural validation of Chain A Carboxylesterase Est2 from *Acinetobacter oleivorans* DR1 was performed using the PROCHECK (Laskowski et al., 1993), ProSA analysis (Wiederstein and Sippl, 2007), and Verify-3D (Bowie et al., 1991).

## Active site prediction and multiple sequence alignment-

Carboxylesterase Est2 is an HSL class hydrolase, this class have a conserved pentapeptide motif ($GX_1SX_2GG$) and a catalytic triad (Hemilä et al., 1994)**;** (De Simone et al., 2000)**.** This was confirmed by the multiple sequence alignment using the NCBI cobalt tool (https://www.ncbi.nlm.nih.gov/tools/cobalt/cobalt.cgi?CMD=Web); (Papadopoulos and Agarwala, 2007) and then it was analysed using the ESpript 3.0 server ( https://espript.ibcp.fr/ESPript/ESPript/); (Gouet et al., 1999) .

## MGL tools and Autodock

The Carboxylesterase Est2 protein was uploaded to the MGL tools, and it was prepared by deletion of water molecules, addition of polar hydrogen and by addition of kollman charges of 20. Ligand in .pdbqt format was upload, then no. of torsions was set to 5 and then macromolecule is converted to the .pdbqt format in the grid option and ligand was set for grid. The grid map was generated by covering the conserved pentapeptide motif ($GX_1SX_2GG$) and conserved catalytic triad (S-D-H) with a spacing of 0.375 Å by using Autogrid 4. The dimensions of the grid box were as follows 42 Å × 74 Å × 44 Å and the centre point coordinates were set as X= 3.884, Y= -4.025, Z= -3.041. for docking purpose Lamarckian genetic algorithm and grid supported energy evaluation method were adopted. The number of total GA runs was increased from 10 to 100 and other parameters were used as default. The conformation with minimum binding energy was selected and further visualized using the chimera (Pettersen et al., 2004) and the discovery studio (Gao and Huang, 2011).

## Computational modelling for enhancement of enzyme activity-

Best docked conformation of the Est2 with DEHP having lowest binding energy was chosen and the amino acids that are interacting in a conformer was visualized with the help of Autodock and discovery studio. 20 amino acids are found to be interacting with the DEHP including the conserved residues of the Esterase enzyme i.e. $GX_1SX_2GG$ and triad S-D-H.

Out of these 20, 7 hydrophilic amino acids are chosen as Asp232, Ser 235, Arg236, His 237, Ser 239, Thr 258, and Glu 300, based on rational criteria, all of they were mutated to Ala, Leu, Ile, and Val as these are less hindered group of hydrophobic amino acids using the rotamer in chimera's structure editing tool. After that we get 28 novel Est2 enzyme which are docked with DEHP as the ligand using the same procedure in Autodock 4.2.

# RESULTS AND DISCUSSION

## Screening of Biodegradation

All the three bacterial strains were capable to grow on the DEHP as the sole carbon source, but the *Acinetobacter oleivorans* DR1 shows fast growth compared to the other strains.

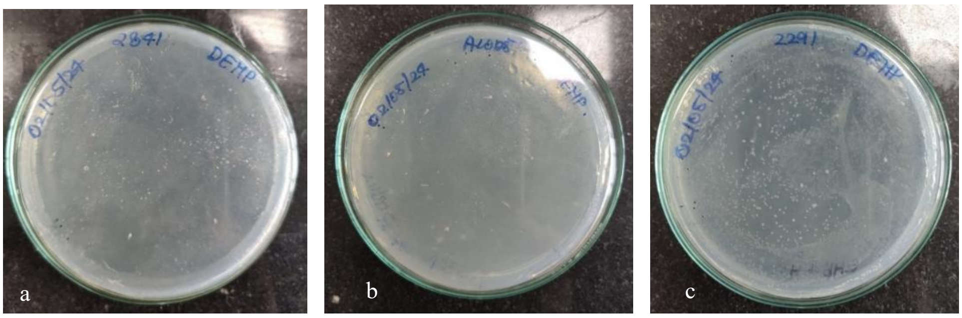
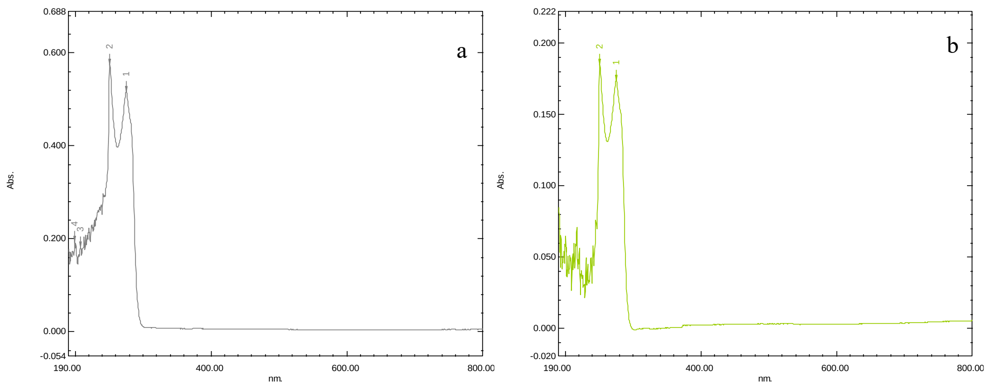



*Figure 11 Plates showing the growth of bacterial strains on DEHP as sole carbon source; a. Arthrobacter sp., b. AWD5, c. Acinetobacter oleivorans* DR1.

## Biodegradation of DEHP

All the bacterial strains after 3 days of incubation were analysed using the UV-Vis spectrophotometry. The results are as follows:



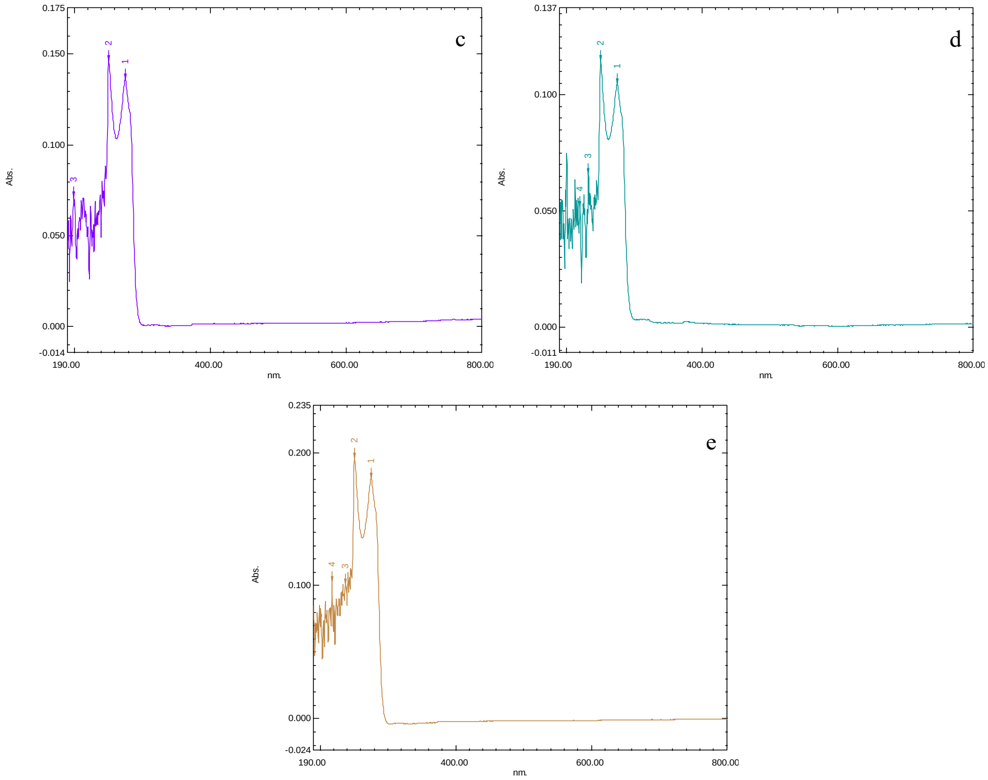



*Figure 12 UV-Vis result of DEHP biodegradation a. Control, b. Klebsiella sp. AWD5, c. Arthrobacter sp. CAS41-1, d. Acinetobacter oleivorans DR1, e. Consortium.*

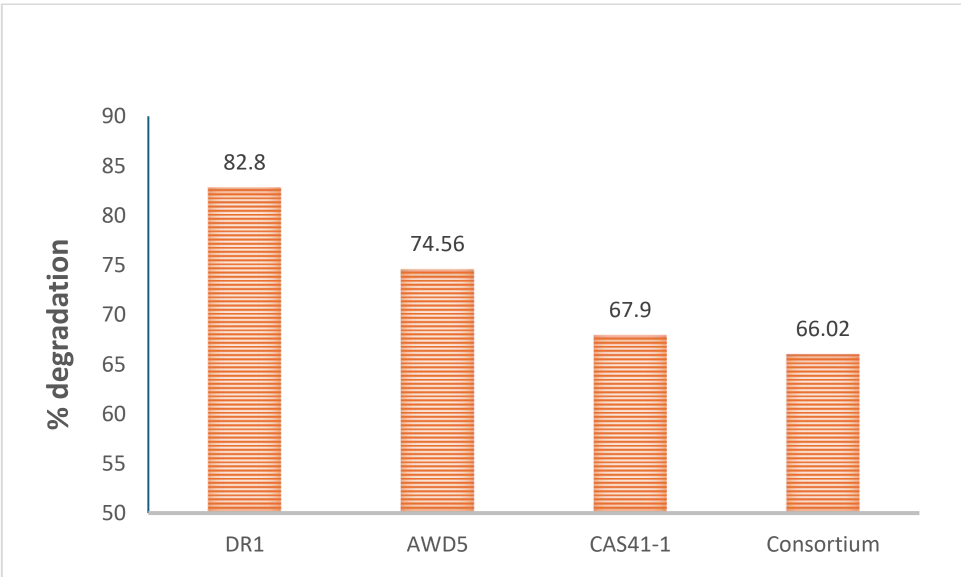



*Figure 13 Biodegradation of DEHP at 72 hrs of incubation.*

Spectrophotometer results showed that the *Acinetobacter oleivorans* DR1 is more capable of degrading the DEHP.

## Congo Red Test

*Acinetobacter oleivorans* DR1 shows positive result for the Congo red, it shows black coloured colonies in the Congo red agar plates. This suggest that the *Acinetobacter oleivorans* DR1 produces extracellular polysaccharide, and it suggests that the *Acinetobacter oleivorans* DR1 strain produces biofilm.

*Figure 14 Congo red test of Acinetobacter oleivorans DR1.*

## Prediction of DEHP degrading protein

From selected gene search in *Acinetobacter oleivorans* DR1 on the JGI-IMG database we found 5 genes with different sequence lengths and varying identities were found which are shown in the Table 2.

*Table 2 JGI-IMG result of the gene search in Acinetobacter oleivorans DR1*

| Genome ID | Query End Coord | Subject Start Coord | Subject End Coord | Bit Score | E-value | Identities | Subject Length | Viral Sequences |
|---|---|---|---|---|---|---|---|---|
| **Chain A, Carboxylesterase Est2** | 10 | 365 | 1 | 356 | 624 | 0.0 | 293/356 82% | 356 |
| **Acetyl-hydrolase** | 120 | 365 | 57 | 294 | 67.0 | 2e-13 | 67/249 27% | 297 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **lipase** | 125 | 339 | 80 | 294 | 87.8 | 2e-20 | 58/215 27% | 324 |
| **acetyl Esterase** | 126 | 349 | 73 | 297 | 105 | 5e-27 | 71/229 31% | 304 |
| **Esterase / lipase** | 131 | 355 | 74 | 295 | 94.7 | 4e-23 | 63/227 28% | 303 |

***Carboxylesterase Est2***

Carboxylesterase Est2 found to be 82% similar with 100% query coverage with the M673 from the *Acinetobacter* M673 having accession no. ADI92748.1.

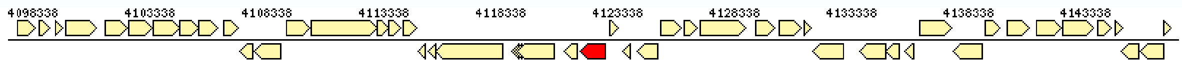



*Figure 15 Gene of the Carboxylesterase Est2 in Acinetobacter oleivorans DR1.*

Est2 was searched on alpha fold for its structure retrieval and a structure with accession no. D8JJC5 was retrieved which is used for MD.

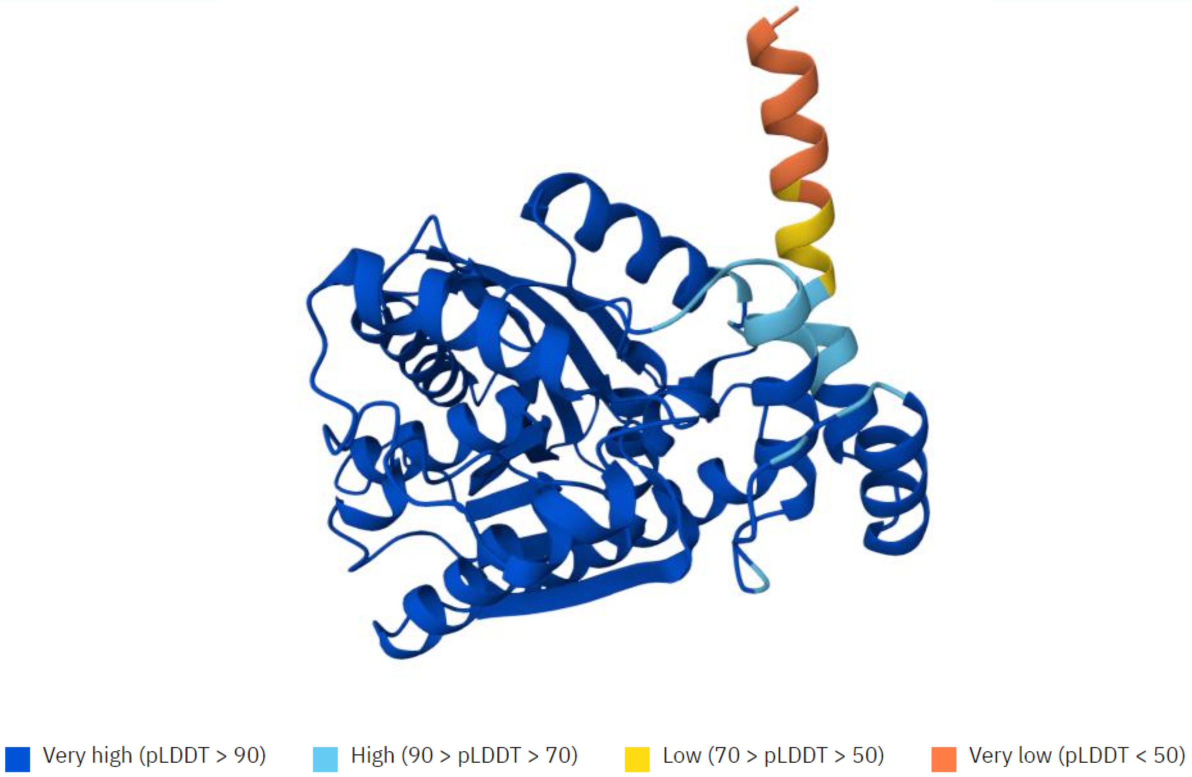



*Figure 16. Alpha Fold structure of Carboxylesterase Est2 and the bar showing structure confidence.*

## Structural validation of Carboxylesterase Est2

PROCHECK analysis of the Est2, produced a Ramachandran plot (Fig 17) which illustrates that the 92% residues are in most favoured regions, 7.7% residues are in additional allowed regions, 0.3% residues are in disallowed regions and no residues are in the generously allowed regions. The results obtained indicate that the protein model is reliable.

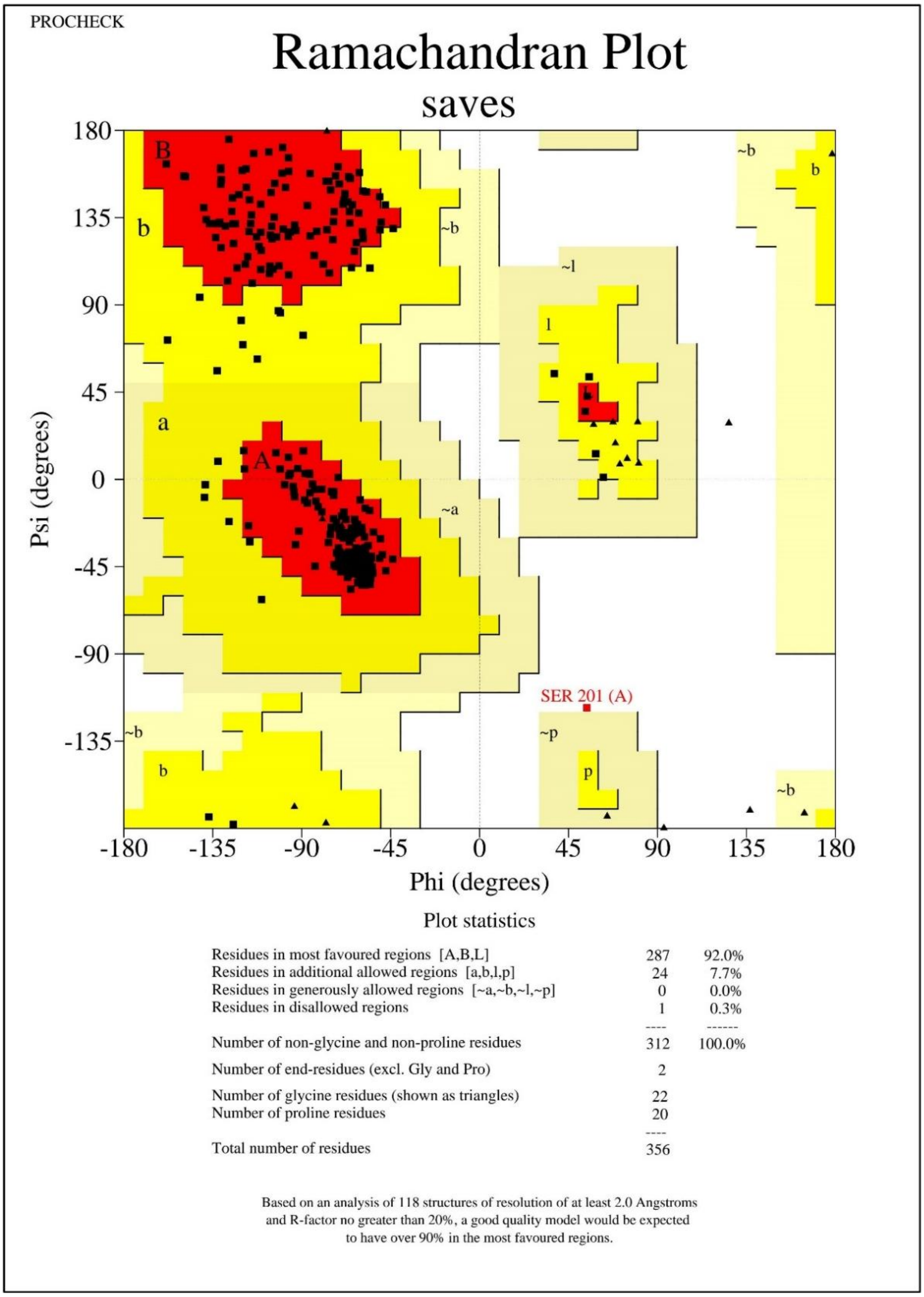



*Figure 17 Ramachandran plot of Carboxylesterase Est2 from Acinetobacter oleivorans DR1.*

ProSA analysis (Fig 18) has further confirmed validity of the structure as the Z score obtained was -8.57. Verify 3D result shows that 83.43% of the residues have averaged 3D-1D score >= 0.1. This result demonstrated that the folding energy pattern of the modelled protein structure is in complete agreement and lies within the expected range.

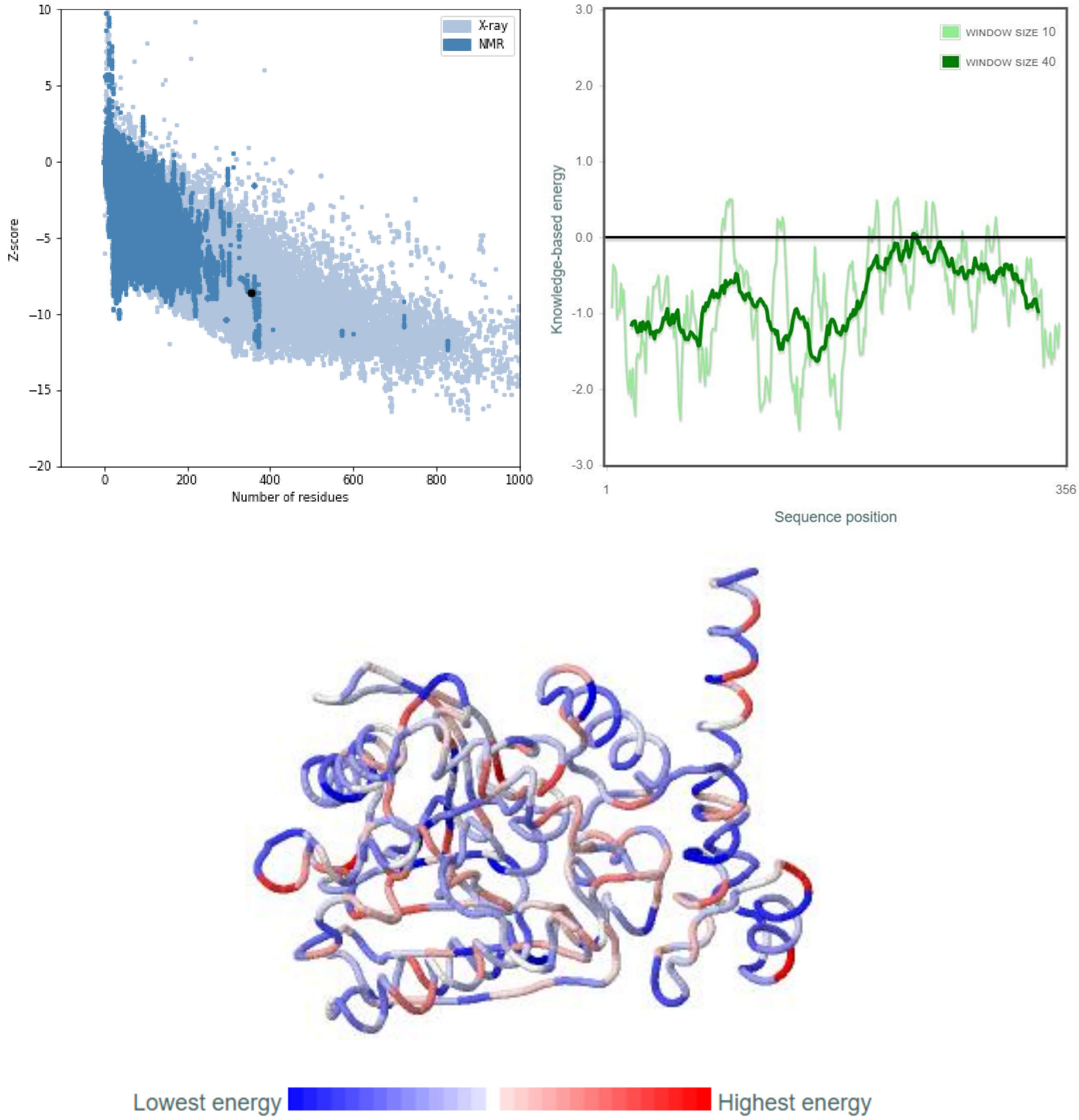



*Figure 18 ProSA analysis of Carboxylesterase Est2 from Acinetobacter oleivorans DR1.*

## Active site prediction-

Multiple sequence alignment (Fig 19) results shows that the Chain A Carboxylesterase Est2 from *Acinetobacter oleivorans* DR1 have the catalytic triad S-D-H and pentapeptide motif ($GX_1SX_2GG$) as in the other hydrolase enzymes. These conserved sequences are chosen as the active site and used for grid formation for the docking studies.

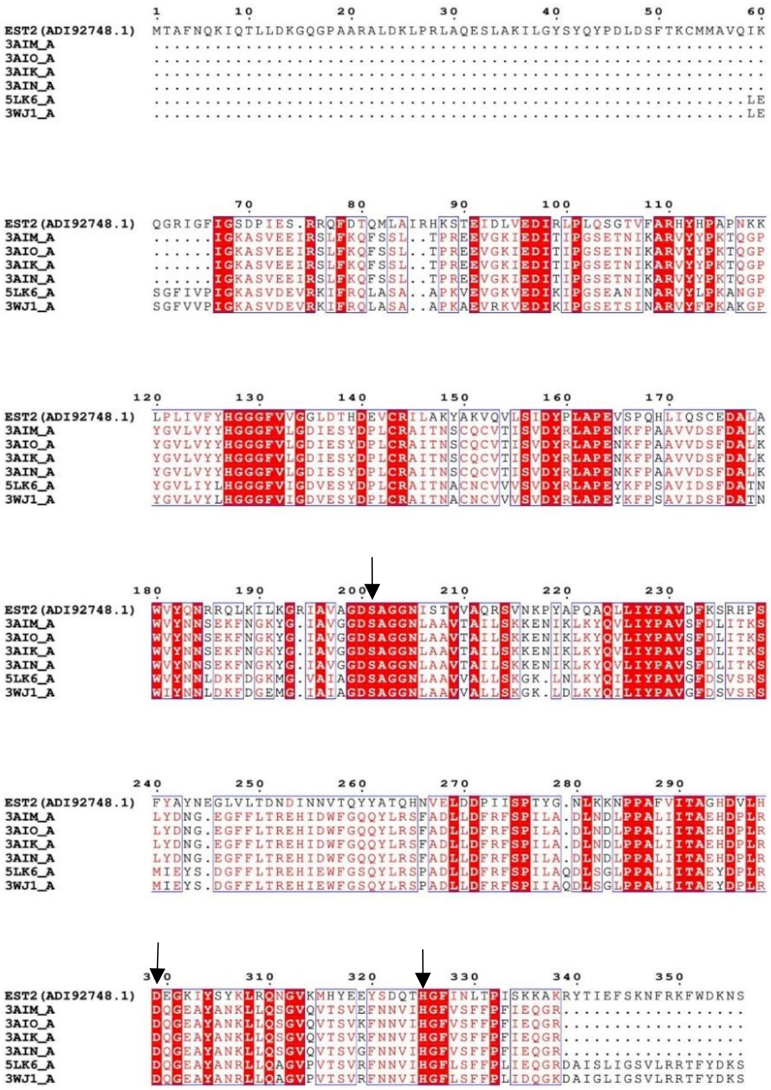


*Figure 19 Multiple sequence alignment of Carboxylesterase Est2 from Acinetobacter oleivorans DR1, arrows showing the conserved regions.*

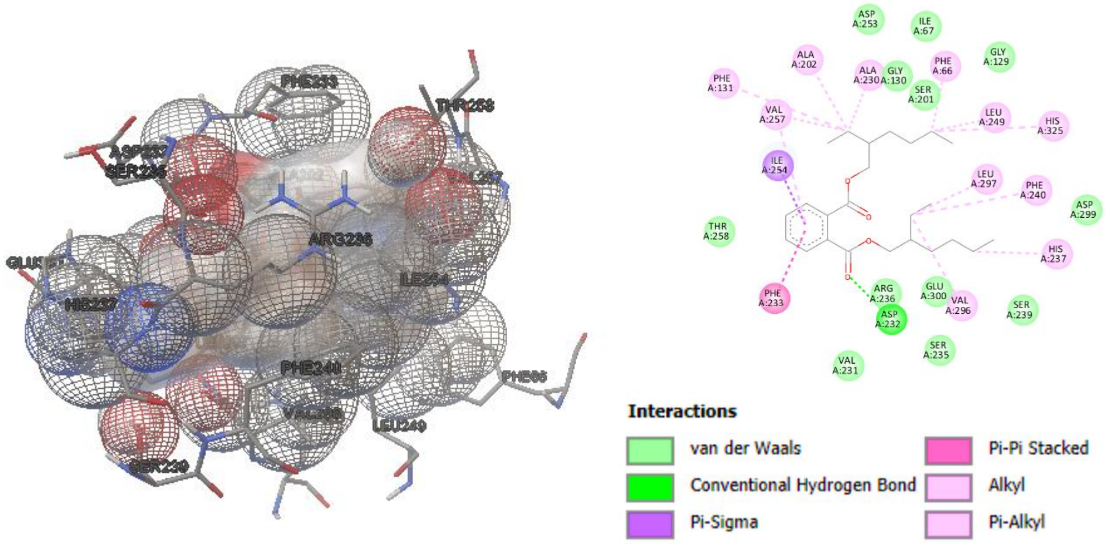



*Figure 20. Interaction site of the Est2 and DEHP*

## Molecular Docking

The docking studies have shown that the DEHP as ligand with the Est2 interacts with a binding energy of -8.09 kcal/mol.

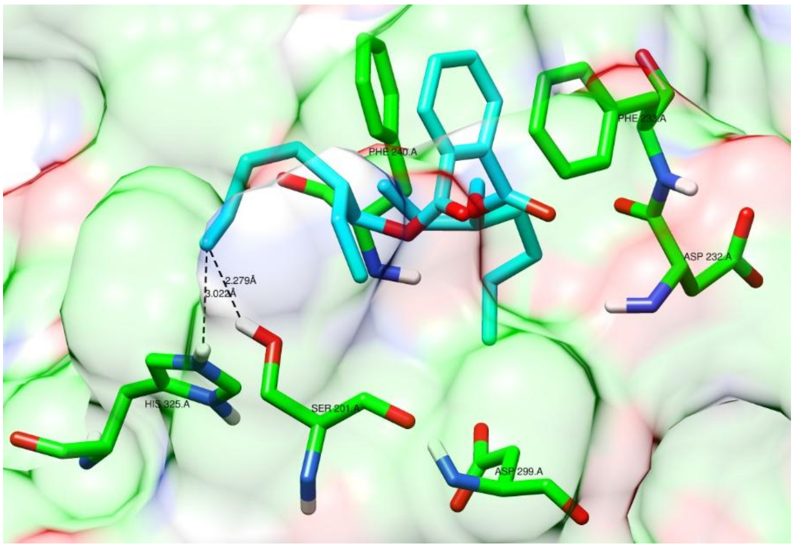



*Figure 21 Binding pocket of the DEHP.*

DEHP is shows to be interacting (Fig. 20) with 20 amino acids which are both hydrophilic and hydrophobic, Phe 66, Gly 130, Ser 201, Ala 202, Ala 230, Asp 232, Phe 233, Ser 235, Arg 236, His 237, Ser 239, Phe 240, Leu 249, Ile 254, Val 257, Thr 258, Val 296, Leu 297, Glu 300, and His 325.

## Computational modelling for enhancement of enzyme activity-

Out of the 20 amino acids found to be interacting with DEHP, 7 amino acids namely Asp 232, Ser 235, Arg 236, His 237, Thr 258, and Glu 300 which are hydrophilic were chosen based on the rational criteria and hydrophilicity, as the other hydrophilic amino acid are conserved.

These 7 amino acids were mutated using the chimera structure editing tool rotamer with the four hydrophobic amino acids Val, Ile, Leu and Ala, and we get 28 novel Est2 enzymes. All these 28 novel Est2 were docked with DEHP as ligand with the same parameter as of the initial dock, with taking the pentapeptide motif and catalytic triad in the grid, this causes some variations in the grid dimensions.

*Table 3 Modification schemes of the Est2 enzyme and scoring functions of the novel Est2 enzymes docking with the DEHP.*

| Novel Est2 Enzyme | Original Amino Acid Residues | Substituted Amino Acid | Binding Energy |
|---|---|---|---|
| Est2 | | | -8.09 |
| Est2-1 | Asp-232 | Ala | -4.21 |
| Est2-2 | | Leu | -5.57 |
| Est2-3 | | Ile | -6.68 |
| Est2-4 | | Val | -4.95 |
| Est2-5 | Ser-235 | Ala | -5.59 |
| Est2-6 | | Leu | -1.18 |
| Est2-7 | | Ile | -6.1 |
| Est2-8 | | Val | -1.13 |
| Est2-9 | Arg-236 | Ala | -7.66 |
| Est2-10 | | Leu | -8.08 |
| Est2-11 | | Ile | -8.7 |
| Est2-12 | | Val | -8.47 |
| Est2-13 | His-237 | Ala | -7.31 |
| Est2-14 | | Leu | 2.05 |
| Est2-15 | | Ile | -7.06 |
| Est2-16 | | Val | -7.18 |
| Est2-17 | Ser-239 | Ala | -5.88 |
| Est2-18 | | Leu | -5.47 |
| Est2-19 | | Ile | -6.38 |
| Est2-20 | | Val | -6.43 |
| Est2-21 | Thr-258 | Ala | -7.87 |
| Est2-22 | | Leu | -5.56 |
| Est2-23 | | Ile | -6.07 |
| Est2-24 | | Val | -6.22 |
| Est2-25 | Glu-300 | Ala | -2.76 |
| Est2-26 | | Leu | -1.9 |
| Est2-27 | | Ile | 2.67 |
| Est2-28 | | Val | 3.8 |

After docking DEHP with novel Est2, we found two novel Est2-11 and Est2-12 enzymes with an increase in the binding energy by -0.6 kcal/mol and -0.38kcal/mol, respectively, which is significant enough to increase the rate of degradation of the DEHP. The interactions pattern in the novel enzymes is same as the wild type as the catalytic triad is found to be interacting with the DEHP and the amino acid mutated also shows an interaction with the DEHP.

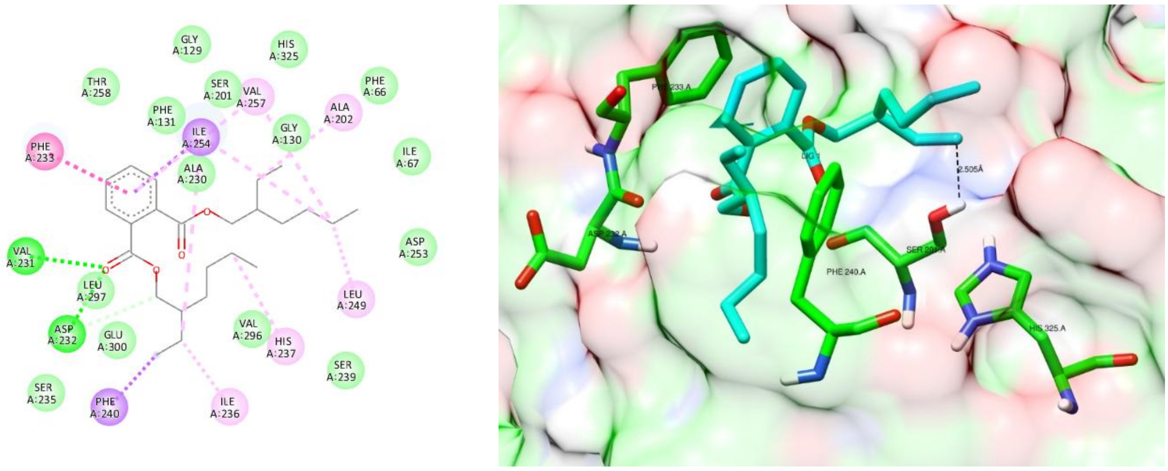



*Figure 22 Novel Est2-11 interactions with DEHP.*

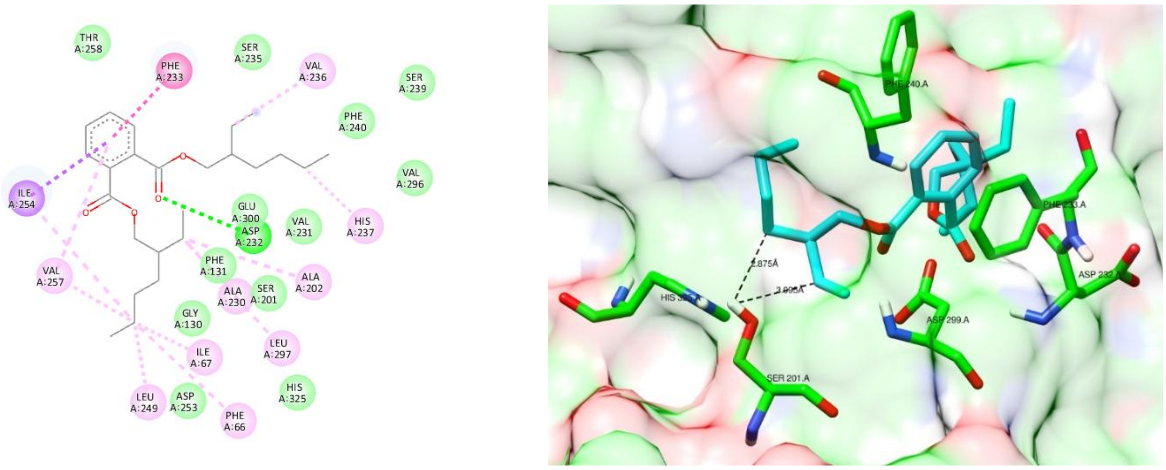



*Figure 23 Novel Est2-12 enzyme interactions with DEHP.*

Also, the sequence alignment results (Fig. 21) further validate the findings, the alignment of Est2 with other Esterase enzymes shows that the other enzymes have either Val or Ile where Est2 have Arg. This evidence further supports the findings.

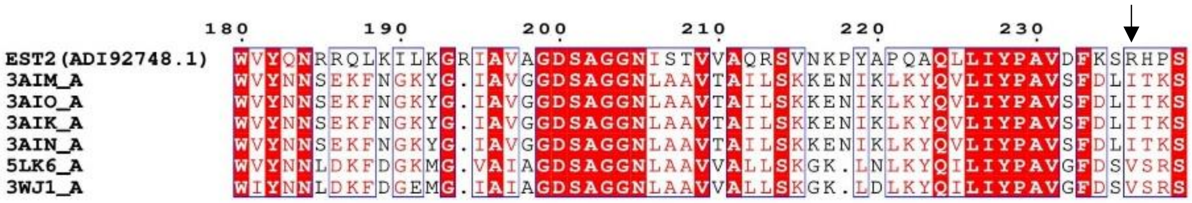



*Figure 24 Figure showing the Arg 236 by an arrow in other Esterase enzymes.*

In this study, *Acinetobacter oleivorans* DR1 was found to degrade DEHP by about 82%, which is quite notable. Cheng et. Al. (2004) reported that *Sphingomonas* sp. DK4 could degrade 90% of DEHP in 5 days, while Feng et al. (2004) found that *Pseudomonas fluorescens* FS-1 managed only a 20% degradation in three days. These results (Table 1) highlight the different abilities of various bacterial species to utilize DEHP as a substrate. Despite multiple studies on DEHP biodegradation, none have specifically targeted *Acinetobacter* species, which are known for their strong oil degrading properties. But Hashizume et al. (2002) studied *Acinetobacter lwoffii* and found it could degrade 20mg/L of DEHP by about 20% in five days.

As people's knowledge and awareness of environmental toxins have grown, biodegradation has gained prominence. On the other hand, not much research has been done on biodegradation using computational methods. The development of quicker and more efficient techniques for cleaning up environmental contaminants like plasticizers and plastics may be facilitated by the integration of computational and experimental research.

In this work, we performed docking studies on the DEHP-degrading enzyme Carboxylesterase Est2, which is derived from *Acinetobacter oleivorans* DR1. The binding energy of -8.09 kcal/mol for DEHP docking with Est2, according to our data, is quite significant for bioremediation. We set out to improve the enzyme's efficiency considering the encouraging results. We used 28 different mutant Est2 enzymes in our docking studies, adopting the methodology of Li et al. (2019). Two of these new enzymes showed better efficiency, as we found that their binding energies rose by -0.6 kcal/mol and -0.38 kcal/mol.

# CONCLUSION

This dissertation has examined the bacterial degradation of Di-(2-ethylhexyl) phthalate (DEHP) and its interactions with Carboxylesterase Est2 through *insilico* methods. DEHP, widely used as a plasticizer in many consumer products, is a persistent environmental contaminant. Its accumulation poses serious health risks, including endocrine disruption and reproductive toxicity, making effective degradation methods crucial.

*Acinetobacter oleivorans* DR1 have shown effectiveness and was able to degrade 82.8% of DEHP under room temperature and neutral pH. For computational analysis the study highlighted Carboxylesterase Est2 from *Acinetobacter oleivorans* DR1 due to its significant similarity to other DEHP-degrading enzymes as one important enzyme involved in the degradation of DEHP. Molecular docking provided insights into enzyme-substrate interactions, identifying key residues involved in the catalytic process. These analyses support Est2's potential as a biocatalyst for DEHP degradation. And further modification of the Carboxylesterase Est2 from *Acinetobacter oleivorans* DR1 shows an increase in efficacy of the enzyme in the degradation process.

This dissertation demonstrates that bioremediation is a viable, eco-friendly approach to addressing DEHP pollution. The identification of effective DEHP-degrading bacteria lays the groundwork for developing biotechnological applications for environmental cleanup. Furthermore, the in-silico studies offer molecular insights that could guide the engineering of more efficient enzymes for bioremediation.

Bacterial and enzymatic degradation of DEHP presents a promising approach to reducing the environmental and health risks associated with this persistent pollutant. Combining microbiological techniques with computational tools can accelerate the development of effective bioremediation strategies, promoting a cleaner and safer environment.

This research advances our understanding of plastic degradation and marks a crucial step towards sustainable management of plastic pollutants. Continued exploration and innovation will help mitigate the negative impacts of plastics on ecosystems and human health. The degradation capabilities of *Acinetobacter oleivorans* DR1 through genetic modifications to improve their efficiency and resilience in various environmental conditions must be undertaken.

This research highlights the biodegradation of DEHP, combining microbiological and computational techniques to develop effective strategies for bioremediation of plastic leach out chemicals and advancing sustainable environmental cleanup.

# FUTURE PROSPECTS

- Validation of enzyme activity and further confirmation of DEHP degradation by GC-MS/MS and HPLC analysis are required
- Enzyme Engineering, use the insights from in-silico analyses to engineer carboxylesterases with greater stability and catalytic efficiency.
- Field applications are required.

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